AEDUINO

Project

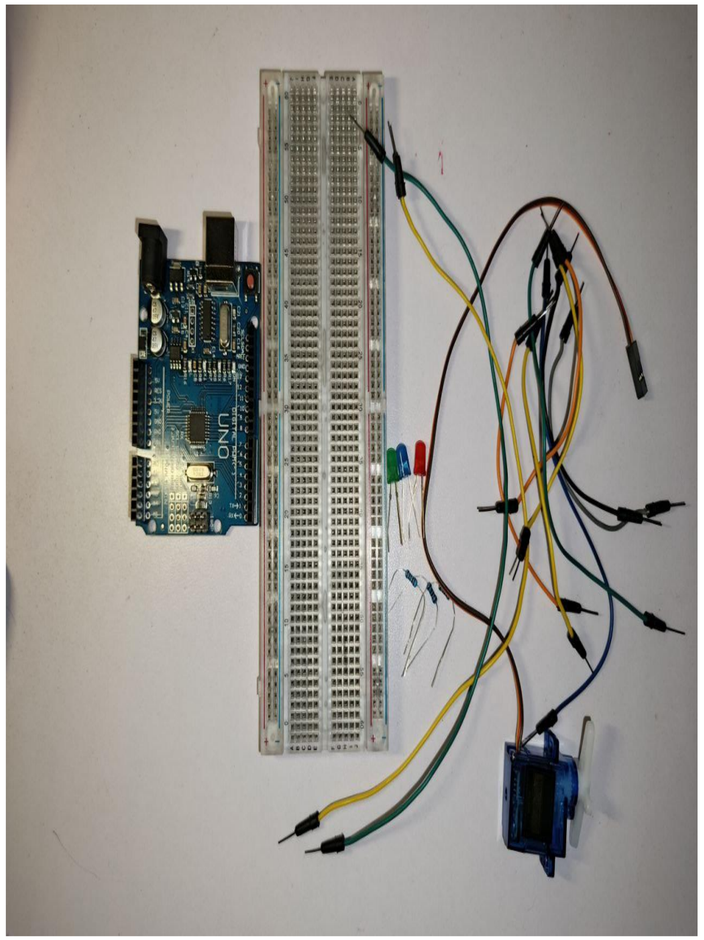
RUPP M4 “2005”



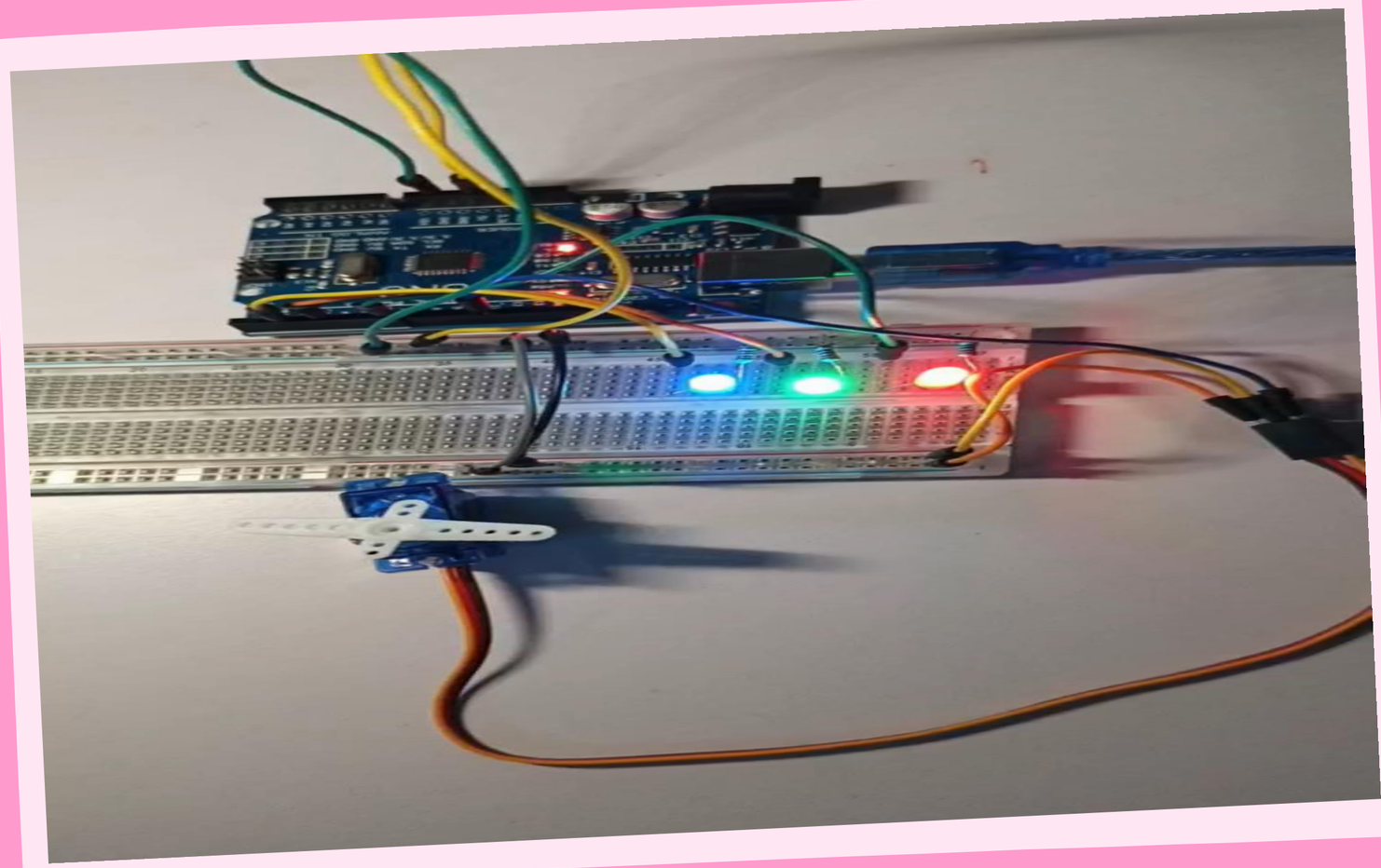
Name Project

- ឧបករណ៍
  - Arduino Board
  - Servo Motor
  - Resistor
  - Breadboard
  - Lcd
  - power supply
- របៀបតឧបករណ៍
  - ភ្ជាប់ជើងGround នៃ Micro Servo ទៅកាន់ជើងGND នៃ UNO
  - ភ្ជាប់ជើង Power ស្ថិតនៅកណ្តាល នៃ Micro Servo ទៅកាន់តង់ស្យូង5V នៃ UNO
  - ភ្ជាប់ជើង Signal នៃ Micro Servo ទៅកាន់ជើងទី11 នៃ UNO
  - ប្រើប្រាស់LCD តាមតម្រូវការការដោយតភ្ជាប់ជើងAnode(ប៉ូលវិជ្ជមាន)ទៅកាន់ជើងណាមួយនៃUNO ហើយភ្ជាប់ជើងCathode(ប៉ូលអវិជ្ជមាន)ទៅជើងGND រួមមួយ

CONCLUSIONS



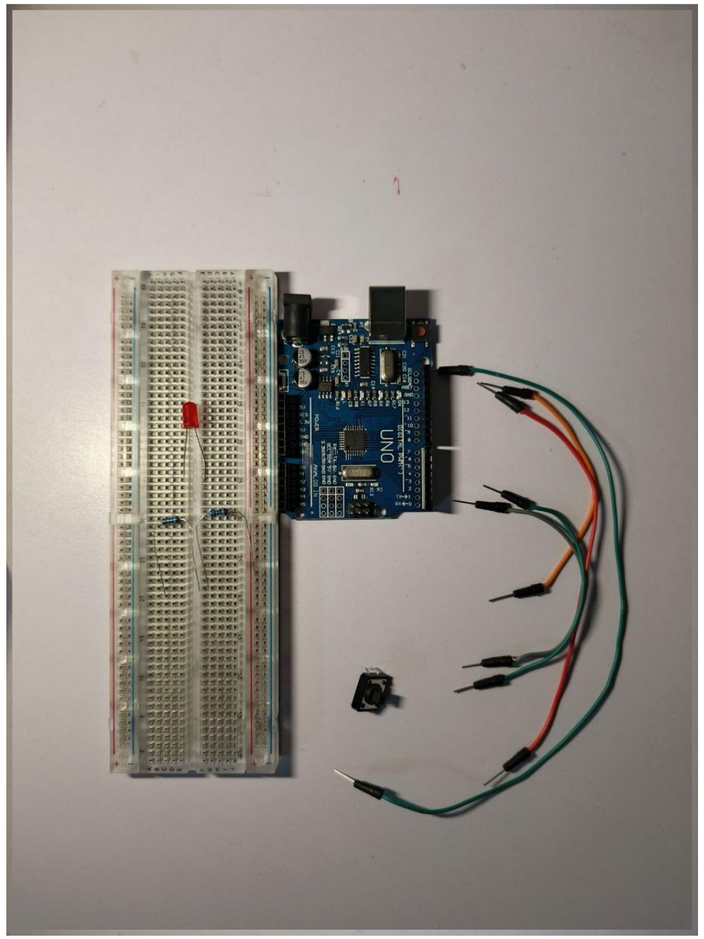
```
#include<Servo.h>
Servo myservo;
void setup()
{
 myservo.attach(11);
  pinMode(7, OUTPUT);
  pinMode(4, OUTPUT);
  pinMode(2, OUTPUT);
}
void loop()
{
  myservo.write(30);
  digitalWrite(7, HIGH);
  delay(1000);
  digitalWrite(7, LOW);
  myservo.write(90);
  digitalWrite(4, HIGH);
  delay(1000);
  digitalWrite(4, LOW);
  myservo.write(180);
  digitalWrite(2, HIGH);
  delay(1000);
  digitalWrite(2, LOW);
  myservo.write(0);
  digitalWrite(7, HIGH);
  digitalWrite(4, HIGH);
  digitalWrite(2, HIGH);
  delay(1000);
}
```



Name Project

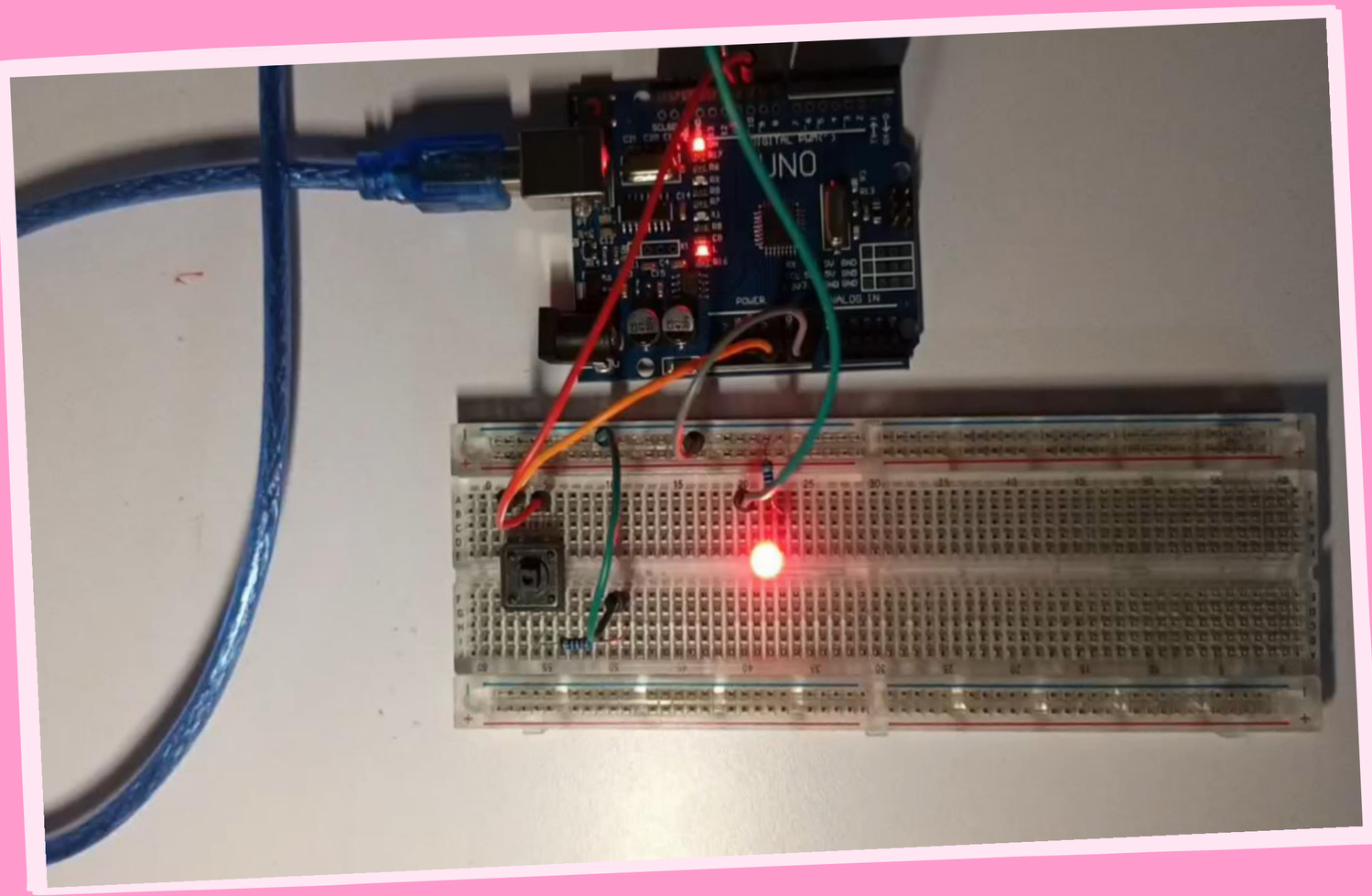
- ឧបករណ៍
  - Arduino Board
  - Pushbutton
  - Resistor
  - Breadboard
  - Lcd
  - power supply
- របៀបតឧបករណ៍
  - កំណត់ជើងណាមួយរបស់ Pushbutton ជាប៉ូលអវិជ្ជមានរួច ភ្ជាប់ជាមួយ Resistor 10 ohm ផ្តុយមកវិញជើងមួយទៀតជាប៉ូលវិជ្ជមាន ត្រូវភ្ជាប់ជាមួយតង់ស្យង5V នៃUNO
  - ជើងមួយទៀតរបស់Pushbutton ដែលមិនទាន់ភ្ជាប់មានលក្ខណៈខ្វែងជាមួយប៉ូលវិជ្ជមាន ត្រូវតភ្ជាប់ជាមួយជើងណាមួយ នៃUNO
  - ប្រើប្រាស់LCD តាមតម្រូវការការដោយតភ្ជាប់ជើងAnode(ប៉ូលវិជ្ជមាន)ទៅកាន់ជើងណាមួយនៃUNO ហើយភ្ជាប់ជើងCathode(ប៉ូលអវិជ្ជមាន) នៃLCD ជាមួយ Resistor បន្ទាប់មកភ្ជាប់ទៅជើងGNDរួមមួយ

CONCLUSIONS




```
#define LED-PIN 11
#define BUTTON-PIN 10
int LED_PIN =11;
int BUTTON_PIN =10;
int state=0;
int value;
void setup()
{
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUTTON_PIN,INPUT);
  Serial.begin(9600);
}

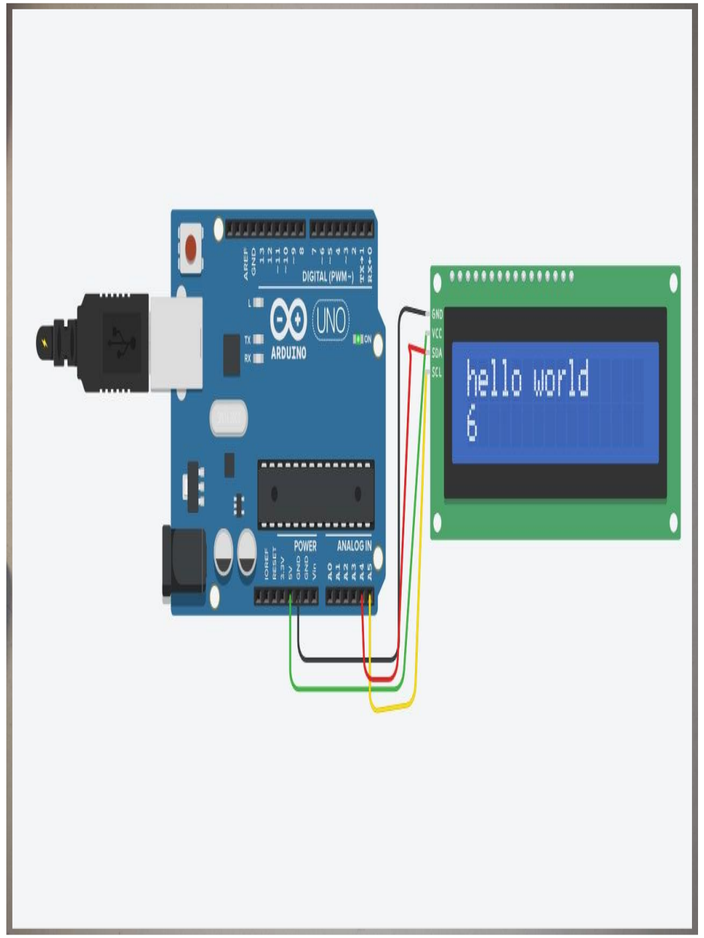
void loop()
{
  value=digitalRead(BUTTON_PIN);
  if(value==1)
    if(state==0){
    Serial.println("No");
    digitalWrite(LED_PIN,1);
    state=1;
  }
  else if(state==1){
    Serial.println("OFF");
    digitalWrite(LED_PIN,0);
    state=0;
  }
  delay(250);
}
```


UNO
POWER
ANALOG IN

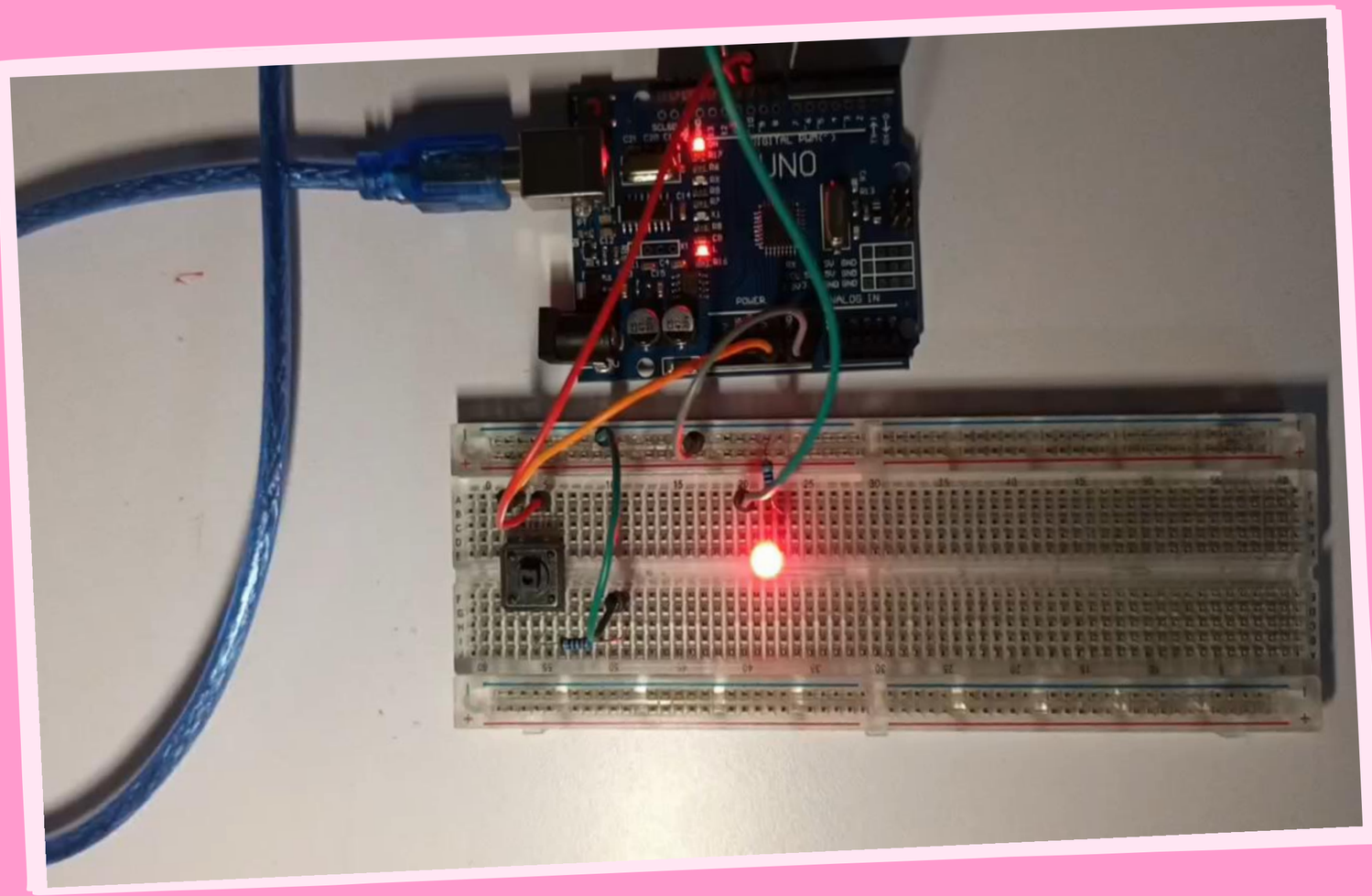
Name Project

- ឧបករណ៍
  - Arduino Board
  - Potentiometer
  - Resistor
  - Breadboard
  - LCD 16×2
  - power supply
- _របៀបតឧបករណ៍
  - បង្កើតជើងតង់ស្យូង5VនិងជើងGNDនៃUNO រួមមួយទៅលើBreadboard
  - ភ្ជាប់Terminal 1 នៃPotentiometer ទៅកាន់ Breadboard មានតង់ស្យូង5V
  - ភ្ជាប់Terminal 2 នៃPotentiometer ទៅកាន់ Breadboard ជើងGND
  - ភ្ជាប់Wiper នៃPotentiometer ទៅកាន់ VO នៃ LCD 16×2
  - ភ្ជាប់ GND និងVCC នៃLCD 16×2 ទៅកាន់ Breadboard ជើងGND និងតង់ស្យូង5V រៀងៗគ្នា
  - ភ្ជាប់RWនិងLED នៃLCD 16×2 ទៅកាន់ Breadboard ជើងGND
  - ភ្ជាប់LED នៃLCD 16×2 ភ្ជាប់គ្នាដោយResistor មួយ ទៅកាន់ តង់ស្យូង5V នៅលើ Breadboard
  - ភ្ជាប់RS, E ,DB4,DB5,DB6,DB7 ភ្ជាប់ទៅកាន់ជើងUNO ទី9,8,7,6,5,4រៀងៗគ្នា

CONCLUSIONS




```
#include<Adafruit_LiquidCrystal.h>
long seconds=0;
Adafruit_LiquidCrystal lcd_1(0);
void setup()
{
  lcd_1.begin(16,2);
  lcd_1.print("hello world");
}
void loop()
{
  lcd_1.setCursor(0,1);
  lcd_1.print(seconds);
  lcd_1.setBacklight(1);
  delay(500);
  lcd_1.setBacklight(0);
  delay(500);
  seconds+=1;
}
```

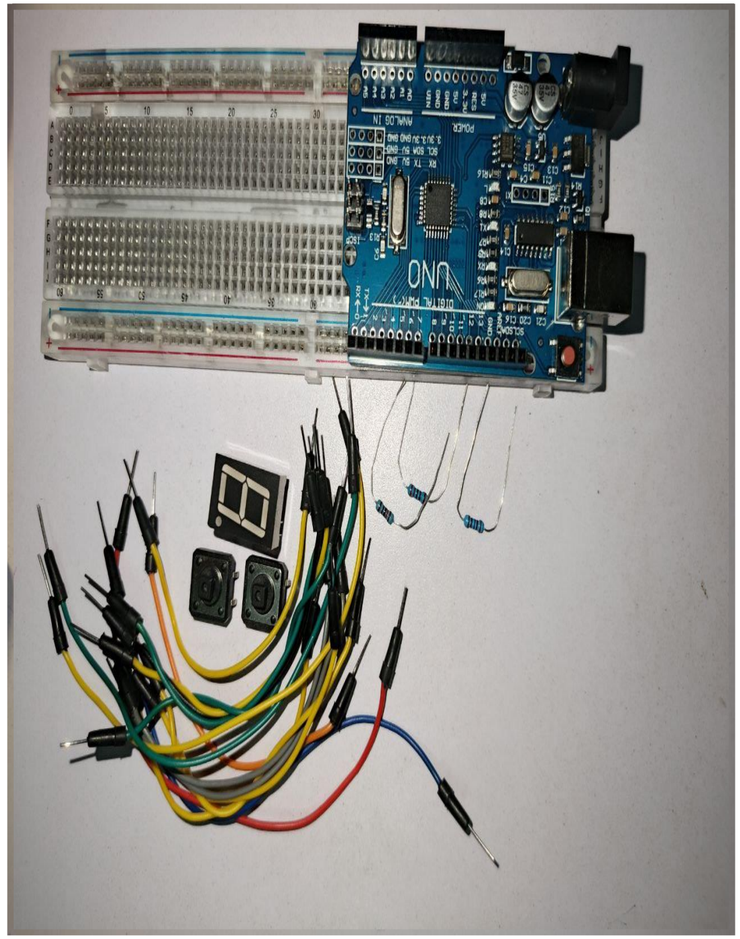

UNO
POWER
ANALOG IN

Name Project

- ឧបករណ៍
  - Arduino Board
  - 7 segment Display
  - Resistor
  - power supply
- បៀបតឧបករណ៍
  - ភ្ជាប់ជើងរបស់ 7 segment ជើង A ,B ,C ,D,E ,F ,G ,DPភ្ជាប់ជានិង Resistor មួយៗ រួចភ្ជាប់ទៅកាន់UNO តាមជើង **7 ,6 ,5 ,11 ,10 ,8 ,9 ,4**

❑ ចំណាំ! យើងក៏អាចភ្ជាប់ទៅកាន់ជើងផ្សេងទៀតក៏បានដែរ រួចគ្រាន់តែយើងបើកPin

CONCLUSIONS

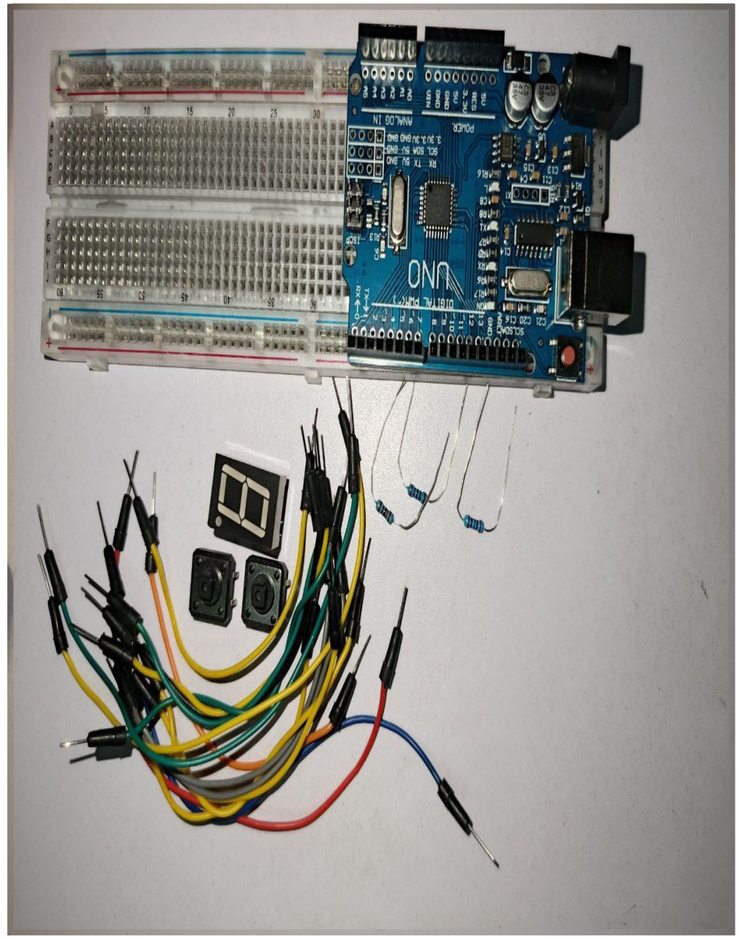



```
int A = 2;
int B = 3;
int C = 4;
int D = 6;
int DP = 5;
int E = 7;
int F = 8;
int G = 9;

int switchUpPin = 13;
int switchDownPin = 12;
int counter = 0;
int buttonUpState = 0;
int lastButtonUpState = 0;
int buttonDownState = 0;
int lastButtonDownState = 0;

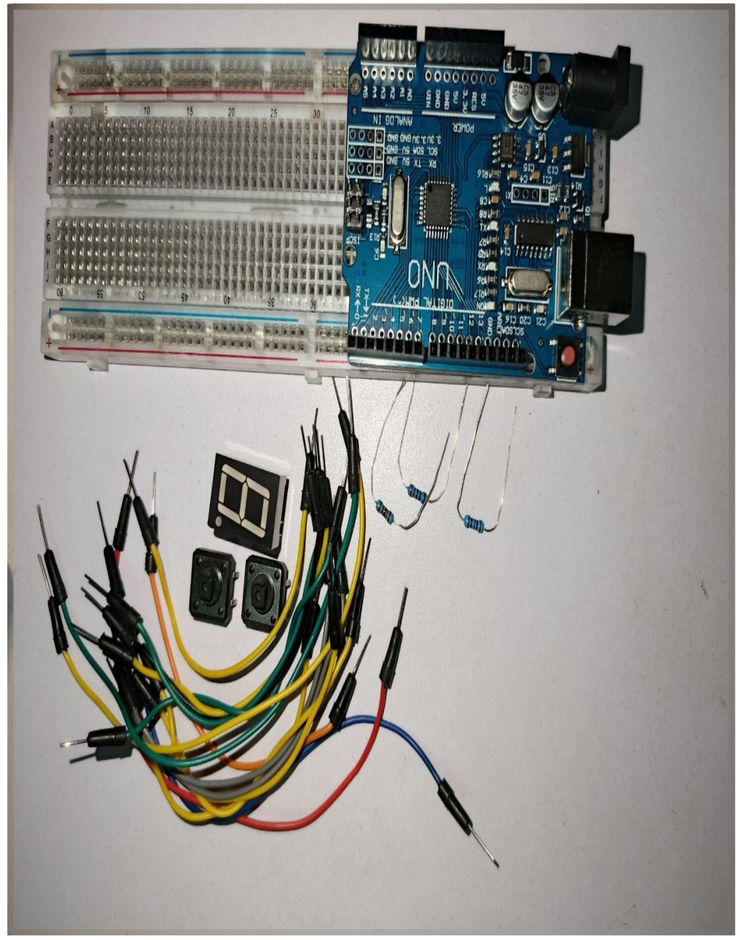
void setup() {
  Serial.begin(9600);
  pinMode(A, OUTPUT);
  pinMode(B, OUTPUT);
  pinMode(C, OUTPUT);
  pinMode(D, OUTPUT);
  pinMode(E, OUTPUT);
  pinMode(F, OUTPUT);
  pinMode(G, OUTPUT);
  pinMode(DP, OUTPUT);
  digitalWrite(DP ,HIGH);
}
```

CONCLUSIONS



```
void loop() {
  buttonUpState = digitalRead(switchUpPin);
  buttonDownState = digitalRead(switchDownPin);
  if (buttonUpState != lastButtonUpState) {
    if (buttonUpState == HIGH) {
      if(counter == 9) {
        counter = 9;
      }
      counter++;
      Serial.println(counter);
      changeNumber(counter);
      delay(300);
    } else {
      Serial.println("OFF");
    }
    delay(50);
  }
  if (buttonDownState != lastButtonDownState) {
    if (buttonDownState == HIGH) {
      if(counter == 0) {
        counter = 0;
      }
      counter--;
      Serial.println(counter);
      changeNumber(counter);
      delay(300);
}
```

CONCLUSIONS

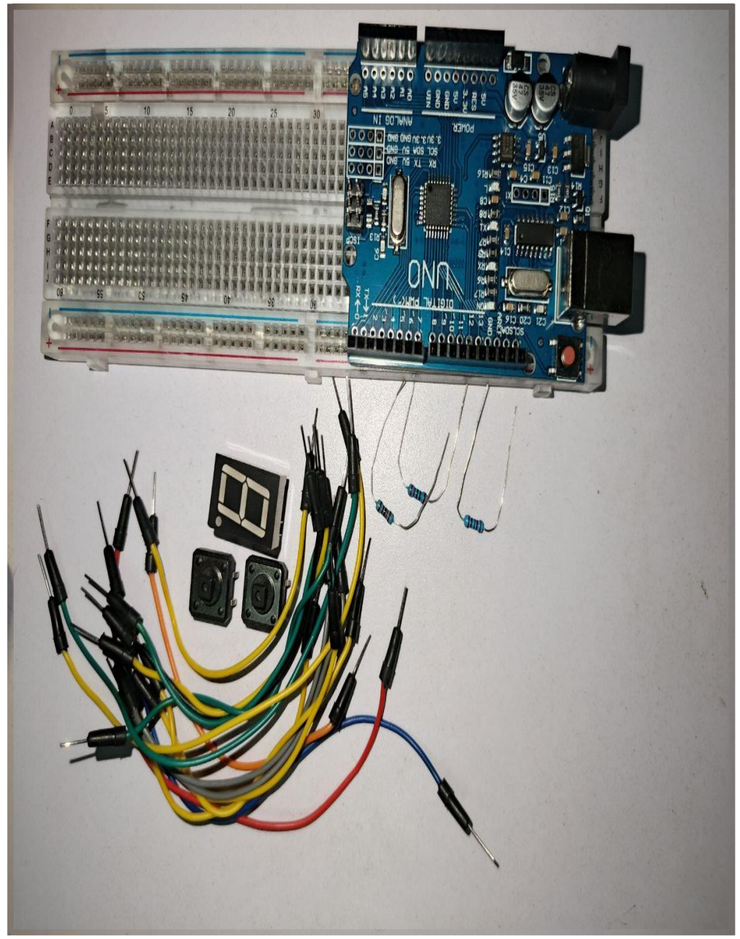



```
else {
     Serial.println("OFF");
   }
   delay(50);
 }
 changeNumber(counter);
}

void changeNumber(int buttonPress) {
 switch (buttonPress) {
   case 0:
   digitalWrite(A, LOW);
   digitalWrite(B, LOW);
   digitalWrite(C, LOW);
   digitalWrite(D, LOW);
   digitalWrite(E, LOW);
   digitalWrite(F, LOW);
   digitalWrite(G, HIGH);
   break;
   case 1:
   digitalWrite(A, HIGH);
   digitalWrite(B, LOW);
   digitalWrite(C, LOW);
   digitalWrite(D, HIGH);
   digitalWrite(E, HIGH);
   digitalWrite(F, HIGH);
   digitalWrite(G, HIGH);
   break;
```

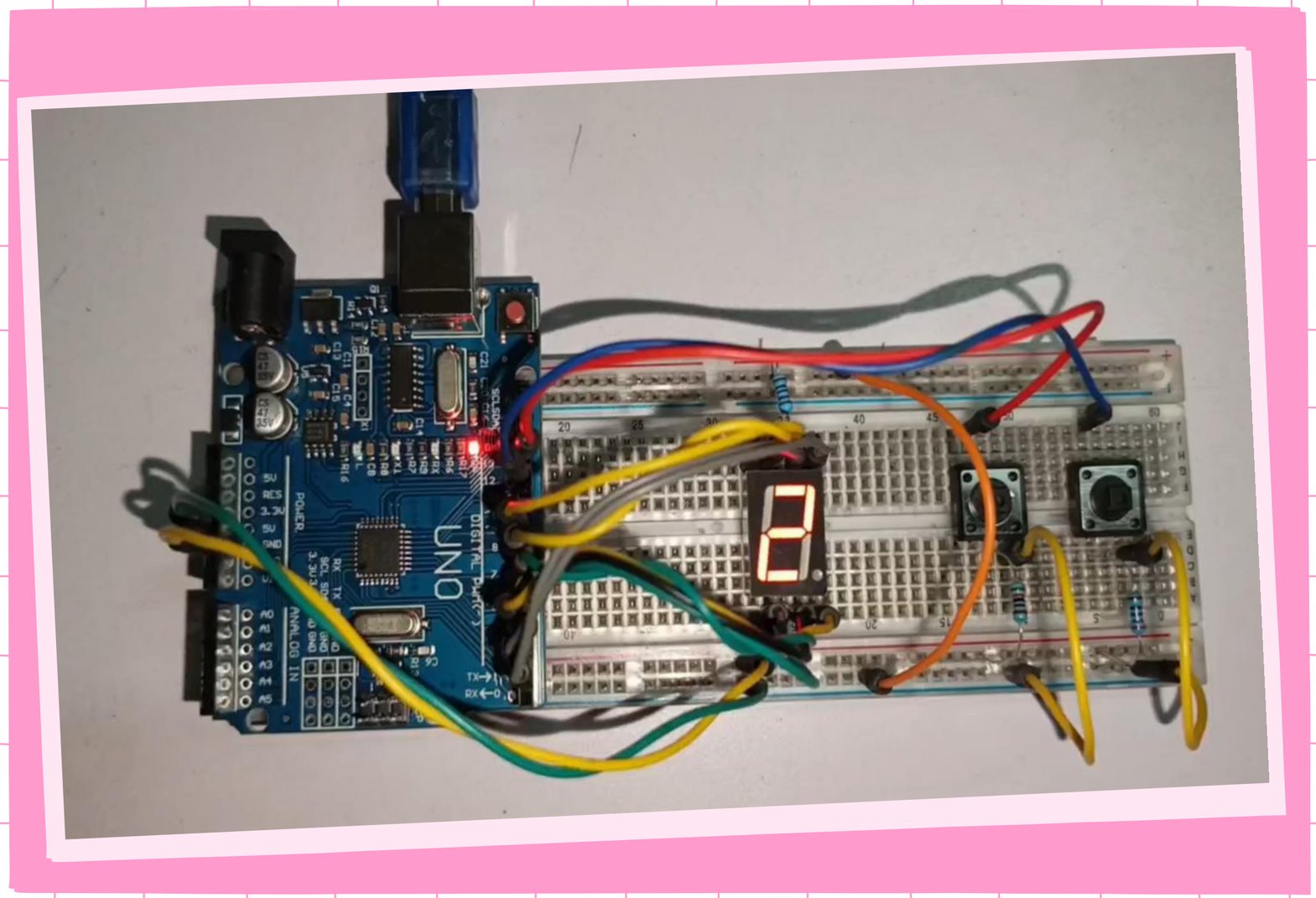
```
case 2:
   digitalWrite(A, LOW);
   digitalWrite(B, LOW);
   digitalWrite(C, HIGH);
   digitalWrite(D, LOW);
   digitalWrite(E, LOW);
   digitalWrite(F, HIGH);
   digitalWrite(G, LOW);
   break;
   case 3:
   digitalWrite(A, LOW);
   digitalWrite(B, LOW);
   digitalWrite(C, LOW);
   digitalWrite(D, LOW);
   digitalWrite(E, HIGH);
   digitalWrite(F, HIGH);
   digitalWrite(G, LOW);
   break;
   case 4:
   digitalWrite(A, HIGH);
   digitalWrite(B, LOW);
   digitalWrite(C, LOW);
   digitalWrite(D, HIGH);
   digitalWrite(E, HIGH);
   digitalWrite(F, LOW);
   digitalWrite(G, LOW);
   break;
```

CONCLUSIONS



```
case 5:
   digitalWrite(A, LOW);
   digitalWrite(B, HIGH);
   digitalWrite(C, LOW);
   digitalWrite(D, LOW);
   digitalWrite(E, HIGH);
   digitalWrite(F, LOW);
   digitalWrite(G, LOW);
   break;
   case 6:
   digitalWrite(A, LOW);
   digitalWrite(B, HIGH);
   digitalWrite(C, LOW);
   digitalWrite(D, LOW);
   digitalWrite(E, LOW);
   digitalWrite(F, LOW);
   digitalWrite(G, LOW);
   break;
   case 7:
   digitalWrite(A, LOW);
   digitalWrite(B, LOW);
   digitalWrite(C, LOW);
   digitalWrite(D, HIGH);
   digitalWrite(E, HIGH);
   digitalWrite(F, HIGH);
   digitalWrite(G, HIGH);
   break;
```

```
case 8:
   digitalWrite(A, LOW);
   digitalWrite(B, LOW);
   digitalWrite(C, LOW);
   digitalWrite(D, LOW);
   digitalWrite(E, LOW);
   digitalWrite(F, LOW);
   digitalWrite(G, LOW);
   break;
   case 9:
   digitalWrite(A, LOW);
   digitalWrite(B, LOW);
   digitalWrite(C, LOW);
   digitalWrite(D, LOW);
   digitalWrite(E, HIGH);
   digitalWrite(F, LOW);
   digitalWrite(G, LOW);
   break;
  }
}
```


UNO
POWER
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Thank you

Contact Us